

Simulated Strawberry Harvesting in NVIDIA Isaac Sim

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Abstract—This project presents a simulated strawberry-harvesting pipeline implemented in NVIDIA Isaac Sim 5.0. The system uses a Franka Panda arm, a parallel gripper, a fixed RGB-D camera, simplified plant models, Lula inverse kinematics, and RRT-based motion planning. For each target, the planner evaluates multiple three-dimensional approach directions and random seeds, selects the path with the shortest estimated grasping time, approaches the strawberry, closes the gripper, removes the fruit, and places it on the ground.

The project designed 10 different trials including changing the position and angle of the strawberry, the angle of the leaf and different positions of obstacle like immature strawberry to validate the algorithm.

I. VIRTUAL HARDWARE AND SIMULATION ENVIRONMENT

The harvesting system was developed in NVIDIA Isaac Sim using a virtual Franka Emika Panda robotic manipulator. The Panda is a seven-degree-of-freedom collaborative robot arm consisting of seven revolute joints.

A parallel two-finger Franka gripper is mounted at the end of the robot. The gripper contains two opposing fingers that move symmetrically to open and close around the strawberry. The seven arm joints are used to determine the position and orientation of the gripper, while the gripper joints control the opening width. In the simulated harvesting sequence, the gripper first remains open during motion planning and approach. After reaching the grasp pose, the fingers close to a predefined width to represent securing the strawberry. The harvested fruit is then logically attached to the gripper and moves together with the end effector during the extraction and placement stages.

II. METHODOLOGY

The harvesting motion-planning system contains five main stages: three-dimensional approach generation, geometric collision pre-screening, inverse-kinematics validation, RRT trajectory planning, and global path comparison.

First, candidate grasp directions are generated around the target strawberry. The planner samples 12 azimuth angles covering the full 360° horizontal range and

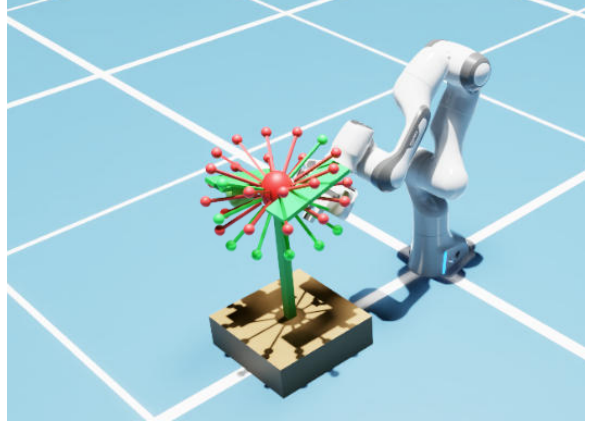


Fig. 1. Route analysis: green lines represent available routes, red lines represent unreachable routes

three elevation angles, producing 36 three-dimensional approach directions. For each direction $\mathbf{d}$, a pre-grasp position is defined approximately 0.08 m from the strawberry center.

Before running RRT, each candidate undergoes geometric collision screening. The segment from the pre-grasp position to the final grasp position is divided into multiple sampled points. A safety volume around the gripper is checked against the leaves, stem, plant base, and neighboring strawberry. Candidates whose final approach intersects an obstacle are rejected.

The remaining poses are tested using inverse kinematics. A candidate is removed if the Franka Panda cannot reach the required position and orientation, if its joint limits are violated, or if the resulting robot configuration is already in collision. The feasible candidates are then ranked by their minimum obstacle clearance, and up to six of the safest approach directions are passed to the RRT planner.

Lula RRT plans a collision-free trajectory from the robot's current joint configuration to each pre-grasp configuration. Because RRT is randomized, several seeds are tested for every approach direction. Different seeds generate different sampling sequences and may produce paths with different lengths and execution times. The seeds also make the results reproducible.

For each feasible trajectory, the program records the

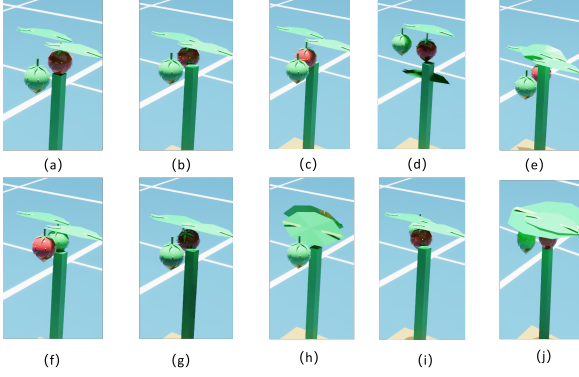


Fig. 2. Ten experimental environments: (a) baseline configuration; (b) left and lower target; (c) right-shifted target; (d) raised target and canopy; (e) low target with side-leaf occlusion; (f) tangent strawberries; (g) tilted target; (h) oppositely tilted leaves; (i) neighboring fruit blocking the preferred approach; and (j) dense overhead canopy.

approach angle, random seed, number of control actions, total joint-space path length, planning wall time, and estimated grasp time. Joint-space path length is calculated as

$$L_q = \sum_{k=1}^{N-1} \|\mathbf{q}_{k+1} - \mathbf{q}_k\|_2.$$

Planning wall time only represents how long the computer takes to generate the path and is not treated as robot execution time. After all feasible paths have been evaluated, they are sorted by estimated grasp time, and only the fastest path is executed.

III. EXPERIMENT

Ten harvesting trials were conducted to evaluate the planner under variations in target position, fruit orientation, leaf occlusion, and neighboring-fruit clearance. The target strawberry was colored red, while the neighboring fruit was colored green. The robot, controller, candidate-generation settings, RRT seeds, and camera configuration remained unchanged across all trials.

Figure 2 presents the ten experimental configurations used to evaluate the strawberry-harvesting system. The red strawberry represents the harvesting target, while the green strawberry is treated as a neighboring obstacle.

IV. RESULTS AND DISCUSSION

Table I summarizes the selected approach, measured wall-clock time, number of actions, and final harvesting outcome for each trial. The selected path is represented as (α, β) , where α and β denote the azimuth and elevation angles in degrees, respectively.

Eight of the ten trials completed the full harvesting sequence, resulting in an overall success rate of 80%. For the successful trials, the average measured time was

TABLE I
SUMMARY OF THE TEN HARVESTING TRIALS.

Trial	Path (α, β)	Time (s)	Actions	Result
1	$(-90, 0)$	51.000	182	Success
2	$(-30, 0)$	56.840	200	Success
3	$(60, 30)$	52.021	168	Success
4	$(-60, 0)$	52.485	162	Success
5	$(-60, 30)$	48.515	164	Success
6	$(60, 30)$	47.280	162	Success
7	$(-30, 0)$	41.789	200	Success
8	$(0, 30)$	—	285	Failure
9	$(60, 30)$	48.595	166	Success
10	$(-30, 0)$	—	210	Failure

49.816 s, and the average trajectory contained 175.5 actions. The mean RRT computation time of the selected paths was 0.0317 s, although this value does not include the evaluation of all alternative approach directions and random seeds.

Trial 8 failed in the environment containing two oppositely tilted leaves. Its selected trajectory contained 285 actions, substantially more than the average of the successful trials. This indicates that the narrow passage forced the manipulator to make a long detour with limited clearance. Although a path to the pre-grasp pose was found, the final approach or post-grasp extraction could not be completed reliably in the restricted space.

Trial 10 failed under the dense overhead canopy. The enlarged leaves restricted both the approach to the target and the space required to remove the strawberry after grasping.

V. CONCLUSION AND FUTURE IMPROVEMENT

The proposed system successfully completed eight of the ten harvesting trials, demonstrating that three-dimensional grasp-candidate generation and RRT-based path planning can support strawberry harvesting under different target positions and moderate obstacle configurations.

Although an RGB-D camera is included in the current simulation, the planner still obtains the positions of the strawberry and obstacles directly from simulator ground truth. Future work will use RGB-D images to detect ripe strawberries and leaves, estimate their three-dimensional positions from depth data, and construct the collision environment automatically. The camera can also provide updated observations during the final approach and fruit extraction, allowing the robot to replan when the target position is inaccurate or the leaves move. This perception-based extension would improve the system's robustness and support its eventual transfer from simulation to a real harvesting environment.